(micro, nano) approach from previous releases. It can be a perfect choice for developers who are fond of Linux with its programming model for drivers, device interfaces and highly-configurable services, all in a single address space.

URL: *www.zephyrproject.org*

Platform.io. This next-generation IDE, with an ecosystem for IoT development, is based on the popular Atom editor with a cross-platform build system and library manager, licensed under Apache 2.0. It wraps popular frameworks like CMSIS, Arduino, mbed, Energia, ST Standard Peripheral Library and WiringPi, and supports native applications on Linux and Windows.

URL: *platformio.org*

Arduino IDE, forks and add-ons. This open source physical computing platform has a simple IDE and coding style. Its power is rapidly getting enhanced by available add-ons for other families of boards, like ESP8266, ESP32 and NRF5x series.

TI Enregia, a fork of Arduino, is available for popular TI launchpads like CC3200, MSP series and Tiva series, with rich support for board peripherals and IoT connectivity. Most of these elements are licensed under GPL 2.

URL: *arduino.cc, energia.nu*

Layer 2

NodeRED. This is a visual tool for wiring hardware peripherals and online services, licensed under Apache 2.0. It has a rich collection of nodes for sensor interfacing, local connectivity (serial, Wi-Fi, Bluetooth, etc), Cloud connectivity (HTTP, mqtt, etc) and social media services. It can run on any OS with Node.js runtime or in a Docker container.

The latest Raspbian for Raspberry Pi and Debian for BeagleBone Black ships with NodeRED, by default. It is also available from Cloud-hosted instances like IBM Bluemix and Sensetecnic FRED. It can even communicate to Arduino-like targets using Firmata protocol. Custom nodes can be built with ease using Node.js backend and HTML frontend. It is a perfect choice for kickstarting IoT prototyping with zero or little programming effort.

URL: *nodered.org, flows.nodered.org*

Eclipse Kura. This is an OSGi based container providing various gateway services addressing M2M and IoT needs, licensed under Eclipse public licence 1.0. Kura components are highly configurable through a Web console, which can be done dynamically. Kura supports good connectivity like CANbus, Serial Bus, Bluetooth LE, Cloud services via MQTT, etc, and even for hardware access using OpenJDK Device I/O library. It aims at reducing the barrier between enterprise and embedded systems with a rich set of Java APIs.

URL: *eclipse.org/kura/*

Macchina.io. This is a toolkit for building embedded applications for IoT on top of POCO C++ libraries and V8 JavaScript engine, licensed under Apache 2.0. The core is implemented in C++ for maximum efficiency and JavaScript is used for application development with its ease of programming. The foundation for macchina.io is Open Service Platform (OSP), which enables dynamically extensible, modular applications based on the powerful plug-in and services model similar to OSGi in Java.

URL: *macchina.io*

Eclipse SmartHome and OpenHAB. Eclipse SmartHome is an OSGi based framework for smart home solutions. It provides a rich set of OSGi bundles for various services and a high degree of modularity.

OpenHAB is a usable product design based on SmartHome framework. It is vendor neutral and hardware/technology agnostic with automation software for rapid development of smart home solutions. Both are licensed under Eclipse public licence 1.0.

URL: *eclipse.org/smarthome, openhab.org*

Iotivity. This Linux Foundation project is initiated by Open Connectivity Foundation (OCF), formerly known as Open Interconenct Consortium (OIC), which comprises a group of companies including Samsung and Intel. OCF aims at seamless device connectivity and standardisation of communications for billions of devices.

A reference implementation of this standard is released with features like powerful device management, resource management, services and security, as well as communication based on CoAP, and has been released under Apache 2.0 licence. This project also provides a few modules for building cloud services for interconnecting Iotivity clients.

URL: *iotivity.org*

Contiki OS and RIOT OS. These operating systems are designed for devices with constrained network and memory resources. These come with full IPv4- and IPv6-capable stacks and support low-power wireless standards such as 6LowPAN and RPL, and protocol connectivity like CoAP. Contiki is licensed under BSD 3-Clause and RIOT under GPL 2.1.

Cooja simulator provided by Contiki allows testing of application developments for large wireless networks with emulation of various hardware targets.

RIOT is a user-friendly OS similar to Contiki and comes with additional offerings like a minimal footprint, C++ APIs, threading and real-time capabilities with minimal low overheads.

URL: *contiki-os.org, riot-os.org*

Note. Even though some level of classification is done based on design goals and primary usage, certain elements are used across layers. For example, Contiki and RIOT can also be used in Layer1, and mbed.org and Zephyr OS can be used in Layer 2, and so on.

Layer 3

Kaa IoT. Kaa is a multipurpose middleware platform for connected